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Interest: Junior Penetration Tester

CodeAlpha internship:

Cyber Security Task # 1

(Basic Network Sniffer)

- Build a Python program to capture network traffic packets.
- Analyze captured packets to understand their structure and content.
- Learn how data flows through the network and the basics of protocols.
- Use libraries like `scapy` or `socket` for packet capturing.
- Display useful information such as source/destination IPs, protocols and payload

Steps:

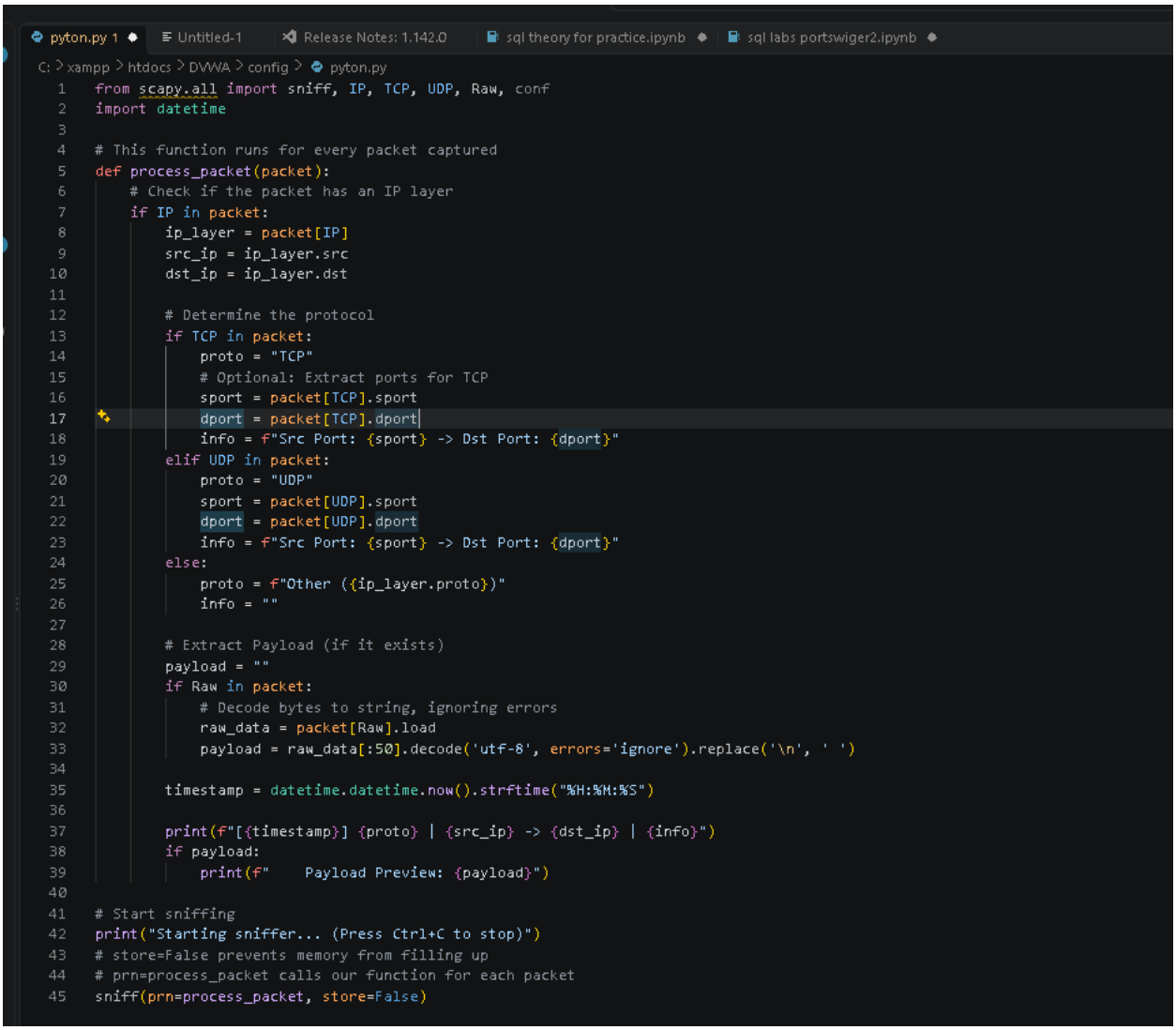
1)**Install Scapy**: Open your terminal and install the library.

sudo apt update

sudo apt install python3-scapy

2)**Verify Installation**: Run python3 -c "from scapy.all import *; print('Scapy OK')" to ensure it is ready .

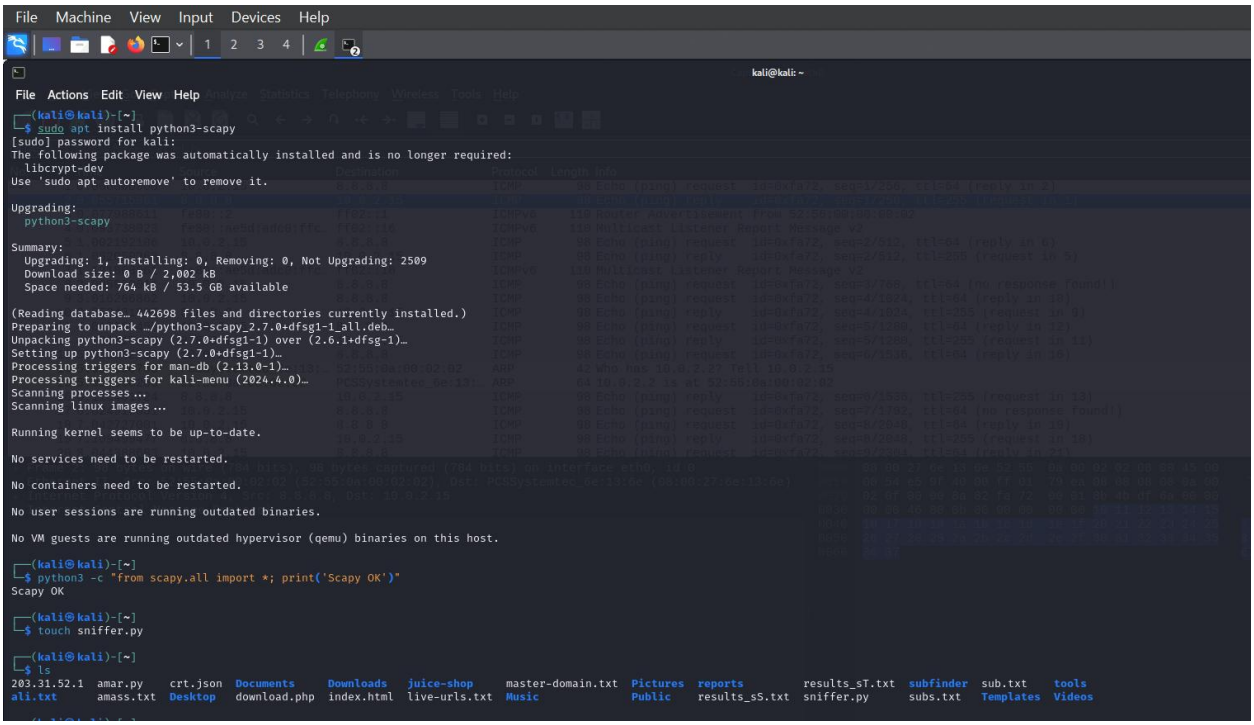
3)Create a file named sniffer.py and write a code here.



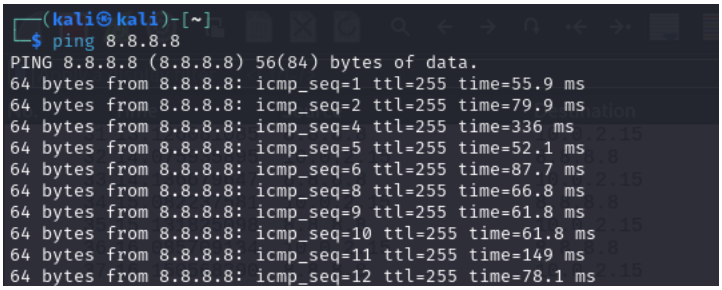
4) **Generate Traffic**: Open a **second terminal** and generate some traffic so your sniffer has something to analyze.

- **Ping (ICMP)**: ping -c 3 8.8.8.8
- **DNS Query**: nslookup google.com
- **HTTP (Unencrypted)**: curl http://example.com

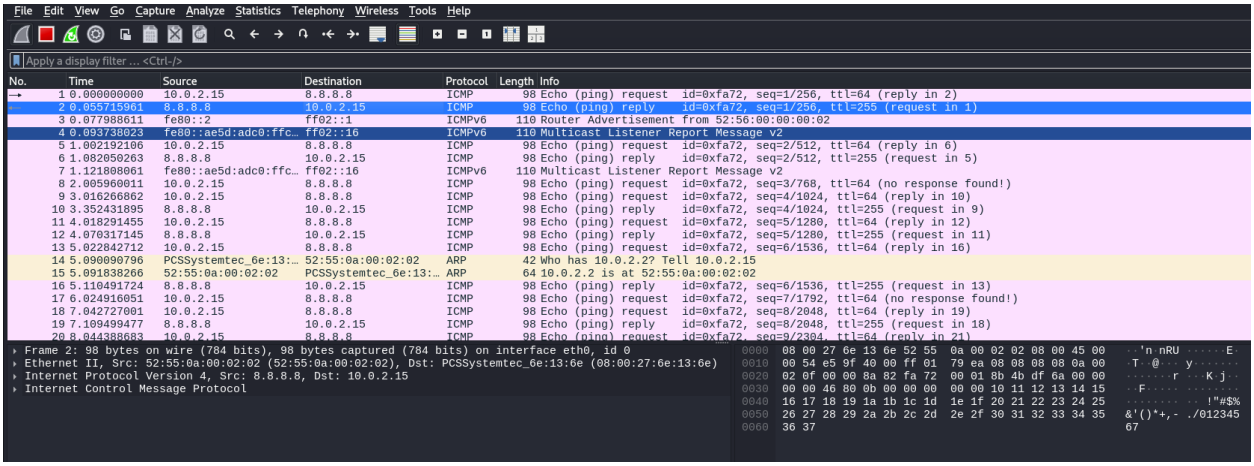
Description:



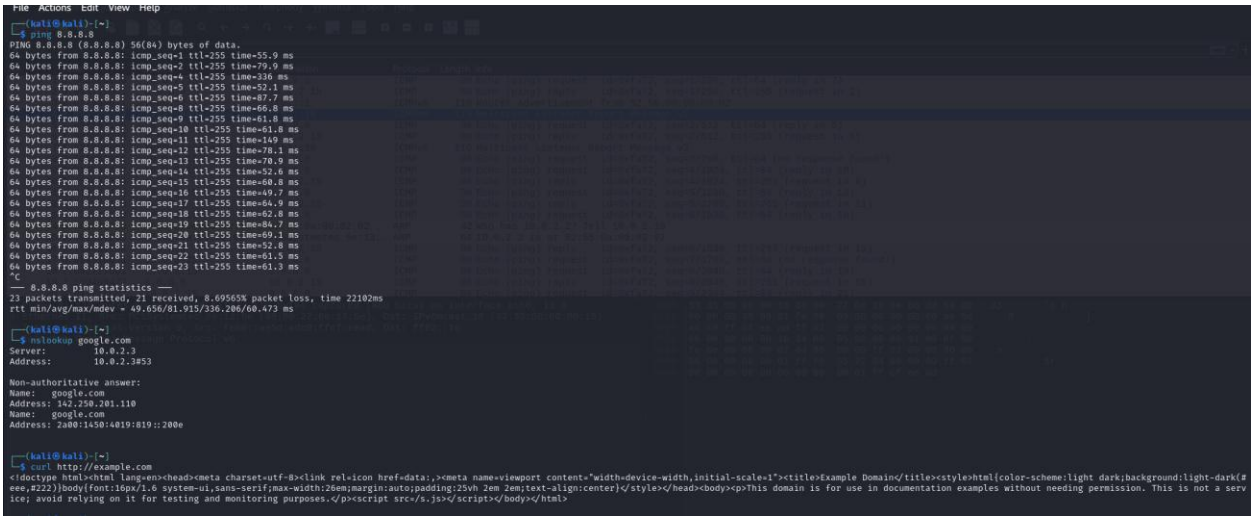
First of all, in this lab I open kali linux and check network flow like ping 8.8.8.8. so, ping 8.8.8.8 is used for check internet connection is good. Other hand, I open wireshark and check their response. Like:



And check response in wireshark:



So, this response is here. Like source is 8.8.8.8 and desination is 10.0.2.15 & protocol and length all of here. So , the request and response check here. What a request how much time and length.

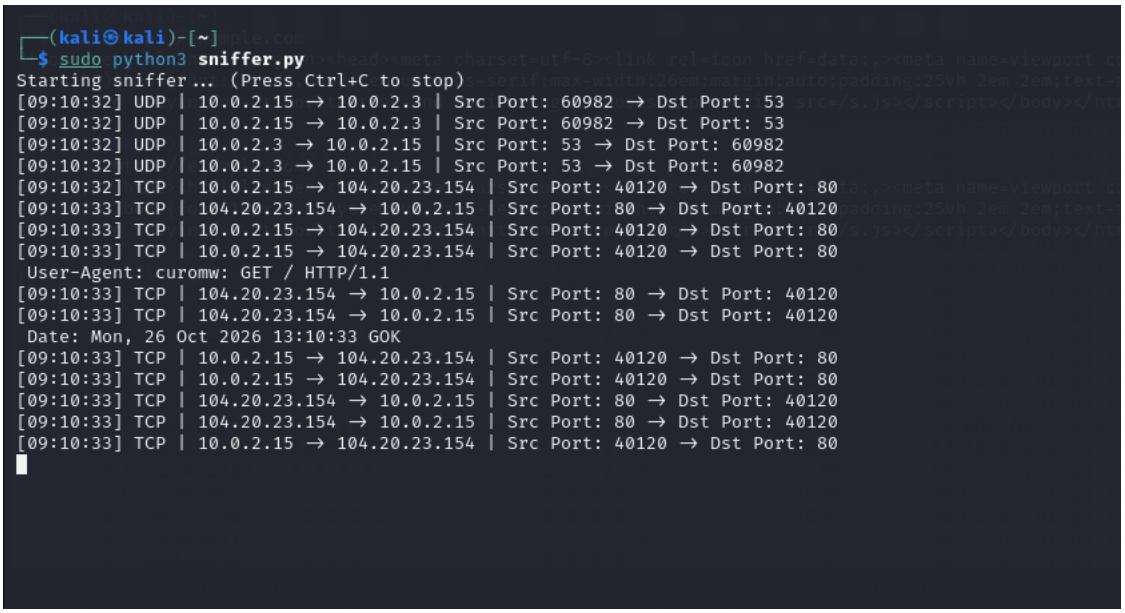


The lab main objective:

So, in this lab not used wireshark, simple put a script and start it.

Like:





The lab is finish: thanks for reading: